Justin Ren Liu

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EDUCATION

University of California, Los Angeles (UCLA) **Expected June 2029**

Bachelor of Science, Electrical Engineering

- Relevant coursework: Signals and Systems, Computer Organization, Logic Design of Digital Systems.

SKILLS & TOOLS

- **Technical Skills:** Multi-layer PCB design, power systems, bare metal firmware, hardware integration, soldering.
- **Tools & Languages:** C/C++, Python, Verilog, Autodesk(PCB/CAD), KiCad, oscilloscope, Git, SolidWorks.

RESEARCH EXPERIENCE

Di Carlo Lab, California NanoSystems Institute at UCLA **August 2026 – Present**

- Independently designed and tuned an ESP32 PI control loop to regulate a cold plate driven by 4 Peltier modules, integrated into eBot Bio's agentic AI platform for autonomous bioassays.
- Reworked the 3D-printed enclosure, lowering minimum cold plate temperature from 2 to -8°C.
- Built complete power harness off a switching supply, with low-side MOSFET switching for Peltier loads.

Mechatronics and Controls Lab, Mechanical and Aerospace Engineering at UCLA **June 2026 – Present**

- Designed the main 2-layer PCB and pressure sensor board for a 6-DOF AUV, with cascaded buck-LDO power regulation with fuses, ESD protection, and (P-Channel MOSFET) reverse-polarity protection, alongside a 9-axis IMU and ESP32–Raspberry Pi compute stack; validated power circuitry in LTspice.
- Worked with mechanical and systems subteams on a 10-person team to integrate electronics into the vehicle, and co-owned the electrical BOM across motors, ESCs, and batteries (sourcing, pricing, and cost reduction).
- Presented in weekly electronics meetings to non-electrical engineers on design updates and tradeoff choices.

PROJECTS

STM32 + FPGA Signal Processing Pipeline **October 2026**

- Implemented STM32F446 peripherals (GPIO, UART, timers, interrupts) via direct register access; achieved 1 kHz ADC sampling using timer-triggered DMA into a 1000-sample buffer with minimal CPU overhead.
- Streamed samples over UART to a Tang Nano 20K, which reassembles them with a Verilog frame parser, filters them, and forwards raw and filtered data to a PC for live Matplotlib plotting.
- Wrote and simulation-verified UART RX/TX modules (baud-rate generator, FSMs, RX input synchronizer).

UCLA Rocket Project Avionics — Valkyrie **November 2025 – June 2026**

- Contributed to the design of multiple 4-layer ESP32-based avionics PCBs featuring LoRa telemetry, GPS, IMU, and dual altimeters for a hybrid rocket achieving ~13,400 ft apogee; assisted in hardware bring-up.
- Helped design and validate avionics power systems for a bodytube PCB (cascaded buck-LDO regulation, ideal-diode ORing), ensuring stable power delivery and seamless USB/battery power switching.
- Soldered and assembled avionics hardware, crimped and soldered board-to-board cable connections for avionics and EGSE systems, and performed debugging and board-level rework to resolve electrical issues.

Vitagotchi — UCLA IdeaHacks 2026 (36h, team of 4) **April 2026**

- Developed a 2x ESP32 wearable system for real-time health state classification (accelerometer, heart rate, temperature/humidity), with ESP-NOW wireless communication, Firebase data logging and web dashboard.
- Designed the desktop companion unit (interactive plush form factor), integrating dual OLED displays, RGB LED, flex sensors, and LLM-generated responses to provide real-time, multimodal user interaction.
- Built portable power systems (boost and LDO regulation) and rapidly prototyped hardware on perfboard.

Custom ESP32 Development Board **February 2026**

- Designed and routed a 2-layer ESP32 development board from component selection to board bring up.
- Routed USB D+/D– as a matched differential pair, placed decoupling capacitors, ground stitching vias, implemented external BOOT and RESET switches and added reverse polarity and ESD protection.

AWARDS

4th Place, ACM AI Mini ML Challenge (UCLA) 2026

- Placed 4th (team of 2, 8h) of 23 teams with a transfer-learning image classifier achieving 98.648% accuracy.